

## Full Xbox Controller Control Map (Surface Code V12)

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### Left Joystick

| Input        | Axis | Function          | Target Motors | Action                             |
|--------------|------|-------------------|---------------|------------------------------------|
| Left Stick Y | 1    | Forward/Backward  | M0, M1        | Both same direction (CW = forward) |
| Left Stick X | 0    | Strafe Left/Right | M2, M3        | Both same direction                |

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### Right Joystick

| Input         | Axis | Function        | Target Motors | Action                                          |
|---------------|------|-----------------|---------------|-------------------------------------------------|
| Right Stick Y | 3    | Pitch Up/Down   | M4, M5        | Opposite direction (M4 CW, M5 CCW = pitch up)   |
| Right Stick X | 2    | Roll Left/Right | M2, M3        | Opposite direction (M2 CW, M3 CCW = roll right) |

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### Triggers

| Input         | Axis | Function  | Target Motors | Action                    |
|---------------|------|-----------|---------------|---------------------------|
| Right Trigger | 5    | Move Up   | M4, M5        | Both CW (same direction)  |
| Left Trigger  | 4    | Move Down | M4, M5        | Both CCW (same direction) |

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### Buttons

| Button | Index | Function  | Action        |
|--------|-------|-----------|---------------|
| LB     | 4     | Yaw Left  | M0 CW, M1 CCW |
| RB     | 5     | Yaw Right | M0 CCW, M1 CW |

| Button | Index | Function           | Action                    |
|--------|-------|--------------------|---------------------------|
| Back   | 6     | Exit Joystick Mode | Ends setupJoyStick() loop |

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## D-Pad (POV Hat)

### D-Pad Direction Command Sent Action

|             |            |                                 |
|-------------|------------|---------------------------------|
| D-Pad Left  | t,angle    | Rotate inner arm left by angle  |
| D-Pad Right | t,-angle   | Rotate inner arm right by angle |
| D-Pad Up    | s,1,+angle | Pitch servo up                  |
| D-Pad Down  | s,1,-angle | Pitch servo down                |

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## Buttons A/B/X/Y

### Button Command Sent Action

|   |            |                            |
|---|------------|----------------------------|
| A | y,-angle   | Rotate outer arm in (down) |
| B | y,+angle   | Rotate outer arm out (up)  |
| X | s,0,+angle | Close claw servo           |
| Y | s,0,-angle | Open claw servo            |

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## Other Console Commands

| Key Input | Command        | Action                      |
|-----------|----------------|-----------------------------|
| t,angle   | Serial from PC | Move inner stepper relative |
| y,angle   | Serial from PC | Move outer stepper relative |
| o         | Serial from PC | Home inner arm (stepper)    |
| y,o       | Serial from PC | Home outer arm (stepper)    |

| Key Input | Command | Action |
|-----------|---------|--------|
|-----------|---------|--------|

|   |  |                     |
|---|--|---------------------|
| j |  | Start joystick mode |
|---|--|---------------------|

|   |  |                        |
|---|--|------------------------|
| q |  | Quit + close Pico port |
|---|--|------------------------|